

The measurement and early warning of daily financial stability index based on XGBoost and SHAP: Evidence from China

Benyan Tan, Ziqi Gan^{*}, Yan Wu

School of Economics and Management, China Three Gorges University, Yichang 443002, China

ARTICLE INFO

JEL Classification:

E44
C52
C53

Keywords:

Financial stability
Daily frequency index
Machine learning
Early warning model
Interpretable framework

ABSTRACT

Financial stability plays an important role in the economic and social development of any economy. This study selects daily frequency data of 20 indicators from the money market, stock market, bond market and foreign exchange market in China as research foundation. Firstly, construct the Index of China's Financial Stability (ICFS) by Dynamic Weighting Method based on the time-varying correlation coefficient, and divide ICFS into three categorical variables (high medium low) by Markov Regime Switching Model, results demonstrate that high risk regime can fully explain the financial risk events occurred in the same period in the history. Secondly, the basic indicators and the categorical variables are put into XGBoost model, it turns out that XGBoost model outperforms other machine learning models in the early warning of financial risk on the performance of various evaluation metrics. Moreover, this study proposed an interpretable framework to give a global and local interpretation of early warning model. It turns out that the risk factors influencing China's financial stability are mainly from real economy, financial institutions, market expectations and real estate market, and we identified the 8 most important features and their early warning values. The results provide significant information to policymakers, market regulators and investors, about employing the early warning values as a useful tool to improve resilience to financial risk.

1. Introduction

Financial risk will seriously damage the foundation of economic operation, maintaining financial stability plays an important role in the overall economic security of an economy. In reality, whether it is the US subprime mortgage crisis, or the European debt crisis all has brought a negative impact on the financial stability in different countries. Mishkin (2011) concludes from the worldwide economic conflagration by late 2008 and early 2009 that the global financial system is far more interconnected and excessive risk-taking that threatened the collapse of the world financial system is far more pervasive than before. As for preventing and defusing financial risks, Grabel (2000) demonstrates that emerging economies facing various distinct types of risks, the project of discovering viable policies to prevent, contain, and ameliorate future crises require an antecedent investigation of the multiple risks facing emerging economies. And Mishkin (2011) also points that

understanding how systemic risk can arise and designing policies to rein in this risk taking are tasks of the highest priority. Similarly, Bisiyas, Flood, Lo, and Valavanis (2012) believe that measurement and analysis are prerequisites for addressing threats to financial stability, they construct a robust framework for monitoring and managing financial stability incorporate both a diversity of perspectives and a continuous process to adapt the continuous changes in systemic risk. It is obvious that the advantages of having a timely picture of the state of the economy are multiple and concern a range of economic actors such as central banks, governmental agencies and private investors, and businesses (Fornaro & Luomaranta, 2020). At present, the structure of the financial system of various countries is more complex and the linkage of financial markets is increasing, how to identify and warn these risk events is crucial to maintaining financial stability. In order to strengthen financial risk management and enhance the effectiveness of financial risk prevention and control, the most urgent task is the identification and early

Abbreviations: ICFS, Index of China's Financial Stability; ML, Machine Learning; SSE, Shanghai Stock Exchange; GARCH, Generalized Autoregressive Conditional Heteroscedasticity; EWMA, Exponentially Weighted Moving Average; SVM, Support Vector Machines; DT, Decision Tree; RF, Random Forest; ANN, Artificial Neural Networks; XGBoost, Extreme Gradient Boosting; SHAP, SHapley Additive exPlanations; PDP, Partial Dependency Plots; LIME, Local Interpretable Model-agnostic Explanations.

^{*} Corresponding author.

E-mail address: gzyyuki@163.com (Z. Gan).

<https://doi.org/10.1016/j.eswa.2023.120375>

Received 11 December 2022; Received in revised form 20 March 2023; Accepted 1 May 2023

Available online 5 May 2023

0957-4174/© 2023 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

warning of financial risks. Construct an index that integrates the information of various financial markets and can measure systemic financial risks relatively accurately, so that financial regulators can correctly monitor the stability of the entire financial market, grasp the current financial risk situation, make regulatory decisions in a more timely and effective manner.

Therefore, it is of great significance to measure Index of China's Financial Stability (ICFS) accurately and timely, identify the status of China's financial stability, formulate a sound early warning mechanism, and analyze the main factors affecting ICFS to prevent and defuse financial risks.

Machine learning (ML) is a novel approach to forecasting that has only been gradually applied to economic analysis in recent decades, and ML offers a set of tools that can usefully summarize various sorts of nonlinear in the data (Varian, 2014). Moreover, the intersection of machine learning with econometrics has become an important research landscape in economics. Machine learning has gained prominence due to the availability of large data sets, no matter in macroeconomics or microeconomics research area (Belloni, Chernozhukov, Fernández-Val, & Hansen, 2017; Goulet Coulombe, Leroux, Stevanovic, & Surprenant, 2022). A key advantage of machine learning models is that they are able to overcome the curse of dimensionality faced by traditional statistical methods due to their superior performance in processing, categorizing, as well as projecting high-dimensional, high-frequency data. And in the modern literature of finance, compared with traditional statistical method, more and more studies found significant evidence of accuracy for prediction and forecasting in machine learning (Ahmed, Atiya, Gayar, & El-Shishiny, 2010; Alessi & Detken, 2018; Beutel, List, & von Schweinitz, 2019; Gogas, Papadimitriou, & Agrapetidou, 2018; Holopainen & Sarlin, 2017; Hu, Tang, Zhang, & Wang, 2018; Ivanková, 2012; Samitis, Kampouris, & Kenourgios, 2020; Tanaka, Kinkyo, & Hamori, 2016).

Many studies are now moving from single algorithms to integrated algorithms, and extreme gradient boosting (XGBoost) is an integrated learning algorithm based on gradient boosting that is now widely used in various fields because of its efficient learning capability. For example, XGBoost is generally used to model default risk in P2P network loans, and the model had high accuracy in default risk prediction (Ma et al., 2018; Zhou, Li, Wang, Ding, & Xia, 2019). Li, Cao, Li, Zhao, and Sun (2020) apply the XGBoost model to the personal loan scenario. However, XGBoost is still less commonly used in the field of financial risk warning. Therefore, in this paper we choose XGBoost to be the main empirical model, and other machine learning models such as Support Vector Machines (SVM), Decision Tree (DT), Random Forest (RF), Artificial Neural Network (ANN) are for robust comparisons. Although machine learning models have good predictive performance, a question that has plagued scholars for a long time is to extract causal interpretations from "black box". One of the directions Financial stability board (FSB) puts forward for financial supervision of machine learning is to improve the interpretability of machine learning technology applications. As a matter of fact, the increasingly complex model will limit the ability of developers and users to fully control and interpret the model, and even lead to their inability to understand the operation mode of the model. With the continuous development of research, a lot of studies (Friedman, 2001; Goldstein, Kapelner, Bleich, & Pitkin, 2015; Gramegna & Giudici, 2021; Joseph, 2019b; Zhao & Hastie, 2021) have contributed to the interpretability of machine learning models.

In order to better carry out risk management, the results of algorithm in this paper also should be "explainable" while developing machine learning technology in financial stability area. Therefore, we applied the interpretable machine learning tools currently widely used, including SHAP and LIME. The main objectives of the paper are to (i) construct the index of China's financial stability that can measure systemic financial risks relatively accurately, (ii) establish a financial risk early warning model, find out the best model through dataset fitting and accuracy evaluation, and analyze the relationship between various risk factors

and risk status, (iii) through the interpretable framework, analyze the marginal contribution of each variable in the early warning model, and investigate the safe point and warning point of each variable when warning financial risks.

The empirical part of the study is structured as shown in Fig. 1.

First, we constructed ICFS with 20 basic indicators by dynamic weighting method based on the time-varying correlation coefficient, and then ICFS was divided into three categorical variables (high, medium and low risk) by Markov regime switching model.

Next, we consider the characteristics of each sub-market, and build an early warning model with 20 basic factors in the money market, stock market, bond market and foreign exchange market as the input features, and the three categorical status of low, medium and high financial risk (0, 1 and 2) as the output labels. Based on XGBoost model, samples were divided into the training set and the test set in a ratio of 7 to 3, subsequently, the accuracy, precision, recall and F1 score are selected and then calculated as a measure of the empirical models' predictive ability, compared with other machine learning models such as decision tree, support vector machine, random forest and artificial neural network.

Finally, in order to enhance the interpretability of the early warning model, interpretable tools such as feature importance, partial dependency plots (PDP), Shapley additive explanations (SHAP) and Local interpretable model-agnostic explanations (LIME) are introduced, aiming at building a comprehensive interpretable framework containing global and local interpretation methods to explain the intrinsic mechanism of the "black-box model".

The main contributions of the paper are as follows. First, existing research has the problem of oversimplification, some studies exclusively regard banking risk as financial risk, but ignore the impact of non-financial institutions. To avoid one-sidedness in the meaning of financial stability, this study takes a comprehensive set of variables from the money market, stock market, bond market and foreign exchange market into account. To a certain extent, this is combined with the actual situation of China's financial system, which may have some guiding significance for future research on financial stability. Second, there is a certain limitation that the low frequency data such as monthly, quarterly and annual research data cannot reflect the status of financial stability timely. This paper uses daily high-frequency data, which can identify the status of financial stability and possible financial risks in a more timely manner. Third, machine learning methods are often used as "black boxes". Neither the causal relationship between input features and output labels nor the importance of input features to prediction results can be obtained. This study proposes an interpretable framework including SHAP value mapping plots, PDP, SHAP force plots and LIME to give a global and local interpretation of the early warning model. Moreover, the benefits and apparent advantages of interpretable framework in machine learning technology are highlighted in the study.

The remainder of this paper is organized as following. Section "Literature review" reviews the relevant literature. In section "Methodology", we describe the methods to construct a composite index, Markov regime switching model, Extreme gradient boosting (XGBoost), Shapley additive explanations (SHAP) and Local interpretable model-agnostic explanations (LIME). We report the research results in section "Empirical results", and validate the results in section "Robustness test". Finally, we make a conclusion about the above results in section "Conclusion".

2. Literature review

Scholars have studied financial stability issues in abundance, and judging from the existing literature, the research on financial stability issues has mainly focused on the following aspects.

2.1. Definition of financial stability

Over 20 years ago, bank of international settlements has already

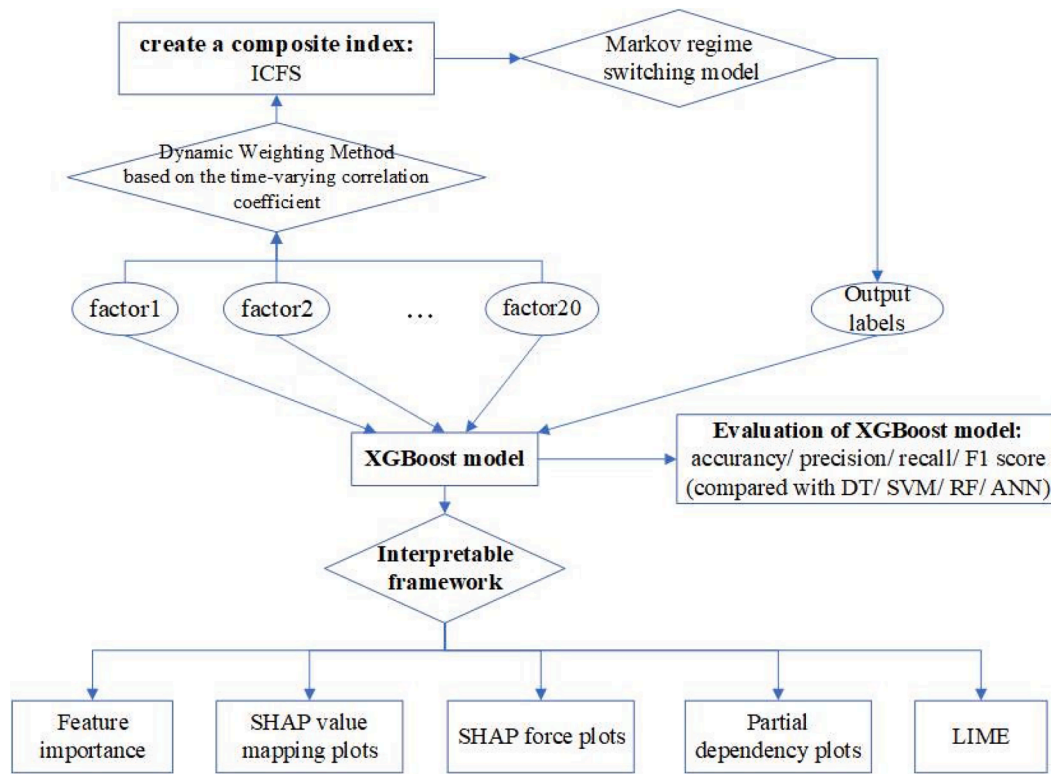


Fig. 1. Visual representation of empirical strategy.

demonstrate the importance of strengthening the safeguards against financial instability (Borio, 2003). In the course of the past research, financial institutions and academic circles have defined the meaning of financial stability from different perspectives. Financial institutions represented by national central banks, such as European Central Bank defines it as a risk of financial instability, as they said “so widespread that it impairs the functioning of a financial system to the point where economic growth and welfare suffer materially”. The Federal Reserve System believes that, financial instability is that the price of financial assets deviates significantly from the basic value, serious distortion of the operation of the financial market and the credit distribution behavior. The International Monetary Fund (IMF) and the World Bank believe that financial stability mainly includes the stability and transparency of macroeconomic policies, prudential supervision of financial institutions, and effective financial infrastructure. The People’s Bank of China defines financial stability as a state in which the financial system can effectively play its key functions.

The academic interpretation of financial stability has also been continuously improved with time. Kaufman and Scott (2003) demonstrate that financial stability refers to systemic stability, rather than the stability of individual financial institutions. Schinasi (2004) defines financial stability as a condition in which an economy’s mechanisms for pricing, allocating, and managing financial risks are functioning well enough to contribute to the performance of the economy. Allen and Wood (2006) conclude that the best approach is to define the characteristics of an episode of financial instability, and to define financial stability as a state of affairs in which episodes of instability are unlikely to occur. Most early definitions of financial stability focused only on the stability characteristics of the financial system itself, but with the rapid development of financial integration, the financial system has become more vulnerable to exogenous shocks, which requires an understanding of financial stability that is not limited to the normal smooth functioning of the financial system, but also focuses on its ability to withstand risk shocks (Deng & Xie, 2021; Houben, Kakes, & Schinasi, 2004; Koong, Law, & Ibrahim, 2017; Petrovska & Mihajlovska, 2013).

It can be seen that although the existing research has different definitions of financial stability, its connotations gradually converge that financial stability refers to that the financial system maintains a stable operation on the one hand, and has the ability to resist risks and external shocks on the other hand.

2.2. Measure of financial stability

Existing measures of financial stability are mainly divided into indirect and direct measures. Indirect measures mainly focus on measuring financial risk, by measuring financial risk to negatively measure financial stability. Indirect measuring methods include Z-score model (Altman, 1968) and SRISK systemic risk measure (Acharya, Engle, & Richardson, 2012; Brownlees & Engle, 2017). In contrast, the direct measuring method is mainly based on the synthesis of indicators, and its logical system is to seek the different dimensions that make up financial stability, and then select the corresponding financial indicators and synthesize them to measure financial stability. For example, an index of financial stress for Canada developed by Illing and Liu (2006), the financial conditions indices for G7 countries proposed by Goodhart and Hofmann (2001), the financial stability index for Colombia constructed by Morales and Estrada (2010), the financial condition indexes for the United States and euro area created by Matheson (2012).

Actually, in 2006, the International Monetary Fund (IMF) published an initiative focused on formulating a definition and single methodology for the compilation of Financial Soundness Indicators (FSIs). Most of the studies are based on the IMF’s FSIs, and the basic indicators that represent the components of the financial system are selected for index synthesis (Gersl & Hermanek, 2007). Due to the fact that the banking sector stands as the sector with the most significant importance within the financial system, most indicators in existing researches are from banking sector.

However, it is obvious that early researches lack of global considerations, as the research gradually deepening, academic gradually adopt comprehensive index method to measure the financial stability, not only

considers the banking system, but also includes the capital market, foreign exchange market, domestic and foreign macroeconomic indicators. For example, [Albulescu \(2010\)](#) selects 20 economic and financial indicators from four dimensions, the development of the financial system, financial vulnerability, financial robustness and the world economic situation, combined them with time-varying weights to build a comprehensive financial stability index. [Arzamasov and Penikas \(2014\)](#) use 16 macroeconomic variables as independent factors by principal component analysis to construct financial stability index for Israel. The dynamic factor model and a broad range of macro-financial variables were adopted to construct a financial stability index to measure the stability of the Malaysian financial system ([Koong et al., 2017](#)). In addition, equal variance weights ([Balakrishnan, Danninger, Elekdag, & Tytell, 2011](#); [Cardarelli, Elekdag, & Lall, 2011](#)), principal component analysis ([Arzamasov & Penikas, 2014](#); [Hakkio & Keeton, 2009](#); [Kliesen & Smith, 2010](#)), time-varying correlation coefficient ([Duprey, Klaus, & Peltonen, 2017](#); [Hollo, Kremer, & Lo Duca, 2012](#)) are commonly used weighting methods in the synthesis of composite index.

In general, the previous academia measuring financial stability methods were basically consistent, including three steps including indicators selection, weight determination and the synthesis of composite index. The difference lies in the selection of basic indicators and the weight determination method.

2.3. Warning of financial stability

Academia has long been concerned about the factors affecting financial stability and conducted early warning analysis. The three classical financial risk early warning models recognized in academia are the FR probability model proposed by [Frankel and Rose \(1996\)](#), the STV cross-section regression model proposed by [Sachs, Tornell, and Velasco \(1996\)](#) and the proposed KLR signal model proposed by [Kaminsky, Lizondo, and Reinhart \(1998\)](#). On the basis of the above three classical models, domestic and foreign scholars have conducted a lot of research to extend the classical models ([Bussiere & Fratzscher, 2006](#); [Cipollini & Kapetanios, 2009](#); [Erdogan, Bennett, & Ozyildirim, 2012](#)). However, with the rapid development of big data and artificial intelligence, the empirical researches paradigm of economics have been profoundly changed, and some risk early warning methods have gradually changed from traditional statistical methods to data-driven machine learning methods. At present, the most common models in the research of the early warning mechanism of machine learning are Support Vector Machine, Neural Networks and Random Forest, etc.

For example, [Nag and Mitra \(1999\)](#) develop a currency crisis early warning system by using an artificial neural network (ANN) model. [Davis and Karim \(2008\)](#) find the binary recursive tree (BRT) technique has good ability to detect interactions between multiple explanatory variables and it's especially applicable to predict large banking crises datasets. [Ahmed et al. \(2010\)](#) present a large scale comparison study for the major machine learning models for time series forecasting, they consider several machine learning models and reveal significant differences between the different methods. [Ivanková \(2012\)](#) applies Support Vector Machine model for the prediction of CISS (Composite Indicator of Systemic Stress), a stability indicator created by the European Central Bank. Similarly, [Gogas et al. \(2018\)](#) choose support vector machines forecasting model to predict bank failures and the model exhibits a 99.22% overall forecasting accuracy. [Alessi and Detken \(2018\)](#) propose an early warning system to predict banking crises by a modern classification tree ensemble technique, the "Random Forest". Similarly, [Tanaka et al. \(2016\)](#) also have argued that random forests may improve early warning predictions in comparison to conventional methods. [Holopainen and Sarlin \(2017\)](#) conduct a horse race of conventional statistical methods and more recent machine learning methods as early-warning models, results show that the conventional statistical approaches are outperformed by more advanced machine learning methods, such as k-nearest neighbors and neural networks, and

particularly by model aggregation approaches through ensemble learning. [Beutel et al. \(2019\)](#) result that machine learning methods often attain a very high in-sample fit, and they are outperformed by the traditional logit approach in recursive out-of-sample evaluations in the prediction of bank crises. [Samitas et al. \(2020\)](#) create a model to predict and forecast the financial contagion risk probability, and they used decision trees, logistic regression classifiers, Support Vector Machines, nearest neighbor classifiers and ensemble classifiers and found the SVM quadratic was the most accurate in all cases.

It is obvious that there are lots of articles on using machine learning to forecast financial risk in current literature, but most of these researches just stop at fitting the data to the model and ignore the analysis of the mechanism of the model output results. As there are still few interpretable models, the practical application of some algorithms is often not trusted because of the lack of explanation of the basis for decision-making, which has led to the previous limitation of the application of artificial intelligence in the financial field. For this reason, this paper applied an interpretable framework for machine learning model, which can better explore the nonlinear relationship between variables in the model, and also identify the risk warning value of each variable in a timely and effective manner.

Therefore, in the process of building the early warning model, this paper prefers to integrate a variety of single weak models and integrated algorithms into the research category to explore the most suitable model, to search for the highest accuracy. This is experience learned from existing literature, and on this basis we also make further improvements is to analyze the interpretability of the most suitable model.

3. Datasets and methodology

3.1. Datasets

This paper takes daily frequency data of 20 indicators from the money market, stock market, bond market and foreign exchange market as the input features, and takes the three categorical variables of ICFS as the output labels. The data covers the period between October 8, 2006 and May 13, 2022, with a total of 3,898 trading days. The data sources are WIND database, CSMAR database, China Central Depository and Clearing and China Foreign Exchange Trade System.

Based on the existing research ([Balakrishnan et al., 2011](#); [Hakkio & Keeton, 2009](#); [Hollo et al., 2012](#); [Kliesen & Smith, 2010](#); [Li & Liang, 2021](#); [Wheelock & Wohar, 2009](#); [Xu & Li, 2017](#); [Yiu, Ho, & Jin, 2010](#); [Zhang & Gao, 2015](#)), the representative financial markets selected in this paper including the money market, bond market, stock market and foreign exchange market. Because the risks faced by financial industry and the real estate industry are quite different from the real sector, in order to combine the reality of China and effectively identify the source, the transmission path and spillover effect of financial risks, this paper divides the stock market into the real sector, the financial sector and the real estate sector. Based on these six sub-markets, we select 20 basic indicators, and [Table 1](#) is the specific explanation of the basic indicators.

This table presents the 20 basic indicators in 6 sub-markets, and lists the name and definition of indicators.

SSE is the abbreviation of Shanghai Stock Exchange.

3.2. Methodology

This section presents the method to constructing a composite index, Markov regime switching model, extreme gradient boosting (XGBoost), Shapley additive explanations (SHAP), local interpretable model-agnostic explanations (LIME).

3.2.1. Method to construct a composite index

A composite indicator is the mathematical combination of individual indicators that represent different dimensions of a concept whose description is the objective of the analysis ([Saisana & Tarantola, 2002](#)).

Table 1

Summary table of sub-markets and basic indicators.

Sub-market	Name of indicators	Definition of indicators
money market	TED	Difference between 3-month CHIBOR yield and Treasury bond yield
	3M_shibor	Real volatility of 3-month SHIBOR
	7D_reporate	7-days repo rate in interbank market
bond market	term_spread	Difference between the 1-year and 10-year Treasury yields
	credit_spread	Difference between 1-year AAA corporate bond and Treasury bond yield
	SETBI_RealVolatility	Real volatility of China bond composite index
stock market (real sector)	INDI_RealVolatility	Real volatility of SSE industrial index
	COMI_RealVolatility	Real volatility of SSE commercial index
	INDI_CMAX	Cumulative maximum loss of SSE industrial index
	COMI_CMAX	Cumulative maximum loss of SSE commercial index
	INDI_illiquidity	Illiquidity of SSE industrial index
	COMI_illiquidity	Illiquidity of SSE commercial index
stock market (financial sector)	180FINI_RealVolatility	Real volatility of 180 financial index
	180FINI_CMAX	Cumulative maximum loss of 180 financial index
	180FINI_illiquidity	Illiquidity of 180 financial Index
stock market (real estate sector)	REI_RealVolatility	Real volatility of real estate index
	REI_CMAX	Cumulative maximum loss of real estate index
	REI_illiquidity	Illiquidity of real estate index
Foreign exchange market	USDCNY	The volatility of USD/CNY exchange rate (central parity)
	1Y_NDF	Volatility of 1-year NDF prices in the offshore market

This study extends the methodology proposed by Hollo et al. (2012). Comprehensively consider the availability of data and the actual situation of China's financial market, this study aggregated six sub-indices into the final composite index (Index of China's Financial Stability, ICFS).

Specifically, the construction of composite index is divided into three steps.

Step 1, the treatment of basic indicators. The volatility of the stock index is calculated by the GARCH (1, 1) model. The GARCH (1, 1) model is calculated as following:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \varepsilon_t \quad (1)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \gamma_1 \sigma_{t-1}^2 \quad (2)$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim N(0, 1) \quad (3)$$

Here y_t is the AR (1) process, ε_t represents residual, $\varepsilon_t \sim N(0, \sigma_t^2)$; σ_{t-1}^2 representing the first-order autoregressive term of the conditional variance of the disturbance term, and ε_{t-1}^2 representing the first-order lag residual square term. It is based on actual calculation that a falling index reflects an increase in volatility, and conversely a decrease in volatility. The cumulative maximum loss is calculated by Eq. (4):

$$CMAX_t = 1 - x_t / \max[x \in x_{t-j}, j = 0, 1, 2, \dots, T] \quad (4)$$

Eq. (4) represents the relative loss rate of the current price versus the stock price peak in the past one year to measure the market risk during the stock decline. In Eq. (4), T representing the moving window for calculating cumulative losses, which is set as the total trading days of the year, usually around 250 days.

Step 2, data standardization. This paper standardizes the 20 basic indicators into new variables between (0, 1) by the empirical cumulative distribution function. The original variables $x_i (i = 1, 2, \dots, 20)$ representing the 20 basic indicators of the 6 sub-markets, each sample is recorded as $x_{i,t} (t = 1, 2, \dots, n)$, and n is the total samples of 3,898. The original variables are firstly arranged in ascending order, $x_{i,[t]} (t = 1, 2, \dots, n)$, and $x_{i,[1]} \leq x_{i,[2]} \leq \dots, x_{i,[r]} \leq \dots \leq x_{i,[n]}$, $[r]$ representing the order of a sample in a group of data, $x_{i,[1]}$ and $x_{i,[t]}$ representing the minimum and maximum value in the same group of data.

$$Z_{i,t} = \begin{cases} \frac{r}{n}, & x_{i,[r]} \leq x_{i,t} < x_{i,[r+1]}, \quad r = 1, 2, \dots, n-1 \\ 1, & x_{i,t} \geq x_{i,[n]} \end{cases} \quad (5)$$

Based on Eq.5, the basic indicators can be transformed into a new form within the range (0, 1], in which the empirical cumulative distribution function measures the proportion of the number of samples in the observed value x_i to the total observed samples, r is the ordinal number of the observed value $x_{i,t}$ in the window period. The order numbers should be averaged if multiple observations are of equal order numbers. In this paper each sub-market $j (j = 1, 2, \dots, 6)$ get an average weight to obtain the stability index of each sub-market $S_{j,t}$, as shown in Eq. (6).

$$S_{j,t} = \frac{1}{n} \sum_{i=1}^n Z_{i,t} \quad (6)$$

Step 3, the calculation process of ICFS. The determination of the weight mainly includes two aspects. (i) Calculate the weight of each submarket by principal component analysis¹. (ii) Weight the stability index of each submarket in the time dimension through the time-varying correlation coefficient method. Dynamic correlations between the sub-market financial stability indexes are helpful to identify the financial risk events on the time series. And the core feature of the weighting method based on dynamic correlation coefficient is to give it a high weight in the period when the correlation coefficient increases and the volatility rises, and to give it a lower weight when the correlation coefficient is low or the market is relatively stable. The dynamic correlation coefficient matrix is calculated as Eq. (7).

$$C_t = \begin{bmatrix} 1 & \rho_{12,t} & \rho_{13,t} & \rho_{14,t} & \rho_{15,t} & \rho_{16,t} \\ \rho_{21,t} & 1 & \rho_{23,t} & \rho_{24,t} & \rho_{25,t} & \rho_{26,t} \\ \rho_{31,t} & \rho_{32,t} & 1 & \rho_{34,t} & \rho_{35,t} & \rho_{36,t} \\ \rho_{41,t} & \rho_{42,t} & \rho_{43,t} & 1 & \rho_{45,t} & \rho_{46,t} \\ \rho_{51,t} & \rho_{52,t} & \rho_{53,t} & \rho_{54,t} & 1 & \rho_{56,t} \\ \rho_{61,t} & \rho_{62,t} & \rho_{63,t} & \rho_{64,t} & \rho_{65,t} & 1 \end{bmatrix} \quad (7)$$

Eq. (7) represents the correlation coefficient matrix composed of the correlation coefficient ($\rho_{ij,t}$), and $\rho_{ij,t}$ representing the correlation coefficient between the financial market i and j . The variance ($\sigma_{i,t}^2$) and the covariance ($\sigma_{ij,t}$) of $\rho_{ij,t}$ are obtained by Exponentially Weighted Moving Average (EWMA)².

$$\sigma_{ij,t} = \lambda \sigma_{ij,t-1} + (1 - \lambda) \tilde{\sigma}_{i,t} \tilde{\sigma}_{j,t} \quad (8)$$

¹ This paper uses principal component analysis to extract the principal components of the six sub-market financial stability indexes ($S_{j,t}$) and then determines the weights (W_i) of them. Money market ($w_1 = 15.47\%$), bond market ($w_2 = 18.28\%$), real sector ($w_3 = 19.91\%$), financial sector ($w_4 = 17.96\%$), real estate sector ($w_5 = 17.83\%$), foreign exchange market ($w_6 = 10.55\%$).

² In this paper the initial variance ($\sigma_{i,0}^2$) and covariance ($\sigma_{ij,0}$) are calculated based on the sample data from October 8, 2006 to December 31, 2008.

$$\sigma_{i,t}^2 = \lambda \sigma_{i,t-1}^2 + (1 - \lambda) \tilde{S}_{i,t}^2 \quad (9)$$

$$\rho_{ij,t} = \frac{\sigma_{ij,t}}{\sigma_{i,t} \sigma_{j,t}} \quad (10)$$

In Eqs. (8) to (10), $i = 1, 2, 3, 4, 5, 6$, $j = 1, 2, 3, 4, 5, 6$, and $i \neq j$, $t = 1, 2, \dots, 3898$, $\tilde{S}_{i,t} (= S_{i,t} - 0.5)$ indicated the difference between the sub-market financial stability index and the theoretical median 0.5. The smoothing index λ indicating the speed of financial information being adjusted, λ values 0.93 according to Hollo et al. (2012) and λ is assumed to remain unchanged throughout the period.

According to above, the ICFS constructed in this paper is described in Eq. (11).

$$ICFS_t = (S_{i,t} \bullet W_i) C_t (S_{i,t} \bullet W_i)' \quad (11)$$

In Eq. (11), $S_{i,t} = (S_{1,t}, \dots, S_{6,t})$ and $W_i = (w_1, \dots, w_6)'$ represent the financial stability index and the weight of each sub-market, and C_t represents the dynamic correlation coefficient matrix.

3.2.2. Markov regime switching model

The index of China's financial stability calculated in this study can only show the overall trend of the financial situation in history, but it cannot specifically measure and determine the risk status. Therefore, the ICFS needs to be divided as a categorical variable to show their financial risk status.

This paper uses the Markov Regime Switching model proposed by Hamilton (1989) to realize the division of risk status. We choose the three regimes Markov regime switching model, to divide ICFS into three categorical variables, low risk, medium risk and high risk, represented by $S_t = \{0, 1, 2\}$.

Firstly, the autoregressive Markov transformation model is established as shown in the Eqs. (12) to (15).

$$ICFS_t = \alpha(s_t) + \beta(s_t)ICFS_{t-1} + \mu_t \quad (12)$$

$$\mu_t \sim N\left(0, \sum (s_t)\right) \quad (13)$$

$$\mu_{s_t} = \mu_1 s_{1t} + \mu_2 s_{2t} + \mu_3 s_{3t} \quad (14)$$

$$\sigma_{s_t}^2 = \sigma_1^2 s_{1t} + \sigma_2^2 s_{2t} + \sigma_3^2 s_{3t} \quad (15)$$

In the Eqs. (12) to (15), $\alpha(s_t)$ is constant term, $\beta(s_t)$ is the coefficient of the ICFS first-order lag term, μ_t is the residual standard deviation, $\mu_t \sim iidN(0, \sum (s_t))$, $\sum (s_t)$ is the variance matrix that varies with regimes s_t . In this model, there are three risk status of low, medium and high in the dynamic change process of the ICFS. From one status to another status, this process is called as the Markov chain process, and the transition probability P can be expressed as follows.

$$P = \begin{bmatrix} p_{00} & p_{01} & p_{02} \\ p_{10} & p_{11} & p_{12} \\ p_{20} & p_{21} & p_{22} \end{bmatrix} \quad (16)$$

$$p_{ij} = Pr(s_{t+1}=j | s_t=i) \quad (17)$$

$$\sum_{j=1}^3 p_{ij} = 1, \forall i, j \in \{1, \dots, N\} \quad (18)$$

In the above Eqs. (16) to (18), p_{ij} represents the transition probability from status i at time $t-1$ to status j at time t , and p_{ij} satisfies the constraints of Eqs. (17) and (18).

3.2.3. Extreme gradient boosting (XGBoost)

Extreme gradient boosting (XGBoost) is created by Chen and Guestrin (2016). The basis of XGBoost is gradient lifting algorithm, which is the representative algorithm of boosting in ensemble algorithm.

The advantage of the ensemble algorithm lies in combining classifiers with poor performance into more performing classifiers and summarizing modeling results from all weak classifiers, resulting in higher efficiency and flexibility than a single model. XGBoost effectively combines the ideas of decision tree and boosting, improving the classification effect by gradually accumulating multiple decision tree models through multiple iterations.

Firstly, the objective function of XGBoost is shown in Eq. (19).

$$obj^{(t)} = l(f_t) + \Omega(f_t) \quad (19)$$

In Eq. (19), f_t representing the t -th tree model, $l(f_t)$ is the loss function in the risk prediction, $\Omega(f_t)$ is the regular term used to reduce overfitting, which can be expressed as Eq. (20).

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2 \quad (20)$$

In Eq. (20), T represents the number of leaf nodes in the t -th decision tree, γ and λ can decide penalty strength together, ω representing the weight value on each leaf node. The prediction results of the model are the weighted sum of all the decision trees, when the t -th iteration is performed, the prediction result can be expressed by Eq. (21).

$$\hat{y}_i^{(t)} = \sum_{k=1}^K f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i), f_k \in F \quad (21)$$

In Eq. (21), $f_t(x_i)$ representing the t -th tree model, F is the decision tree space, is also the set of all sample risk prediction decision trees. Here $\hat{y}_i^{(t)}$ represents the prediction results of the sample i after the t -th iteration, and $\hat{y}_i^{(t-1)}$ represents the prediction results of the previous $t-1$ trees.

The prediction accuracy of any model is determined by the self-dependent bias and the variance of the model itself, while bias is determined by a specific loss function and variance is affected by the regularity term. Therefore, the objective function can be expressed as the formula for Eq. (22):

$$obj^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) + constant \quad (22)$$

$l(\bullet)$ is the loss function, $\hat{y}_i^{(t-1)}$ representing the prediction results of the previous $t-1$ trees, which can be regarded as a constant, $f_t(x_i)$ is the predicted value of the model that needs to be solved, $\Omega(f_t)$ representing the regularization term.

Finally, the sample classification should be determined. When the objective function achieves the minimum value, the final predicted value of the model prediction $\hat{y}_i^{(t)}$ can be obtained.

3.2.4. Shapley additive explanations (SHAP)

SHAP (SHapley Additive explanation) value, proposed by Lundberg and Lee (2017) adapting a concept coming from game theory Shapley (1953), is a unified measure of feature importance. In this framework, the variability of the predictions is divided among the available covariates, so the contribution of each explanatory variable to each point prediction can be assessed regardless of the underlying model (Joseph, 2019a). In SHAP, the contribution of each feature (ϕ_i) on the model output $v(N)$ is allocated based on their marginal contribution. Shapley values are represented by Eq. (23):

$$\phi_i = \sum_{S \subseteq N} \frac{|S|!(M - |S| - 1)!}{M!} [v(S \cup \{i\}) - v(S)] \quad (23)$$

Where f is the model, M is the number of input feature, N is the set of all input features. The quantity $v(S \cup \{i\}) - v(S)$ expresses, for each single prediction, the deviation of Shapley values from their mean: the contribution of the i -th variable. Intuitively, Shapley values are an explanatory model that locally approximates the original model, for a given variable, with the property that, whenever a variable is equal to

zero, so is the Shapley value; and that if in a different model the contribution of a variable is higher, so will be the corresponding Shapley value (Gramegna & Giudici, 2021).

3.2.5. Local interpretable model-agnostic explanations (LIME)

LIME (Local Interpretable Model-agnostic Explanations) proposed by Ribeiro, Singh, and Guestrin (2016) is a post-hoc model-agnostic explanation technique which aims at approximating any black box machine learning model with a local, interpretable model to explain each individual prediction. What LIME does is trying to fit a local model using sample data points that are similar to the observation being explained. The local model can be from the class of interpretable models such as linear models, decision trees, etc (Gramegna & Giudici, 2021). the explanation produced by LIME is obtained by Eq. (24):

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (24)$$

where G is the class of potentially interpretable models such as linear models and decision trees.

$g \in G$: An explanation considered as a model.

$f: \mathbb{R}^d \rightarrow \mathbb{R}$.

$\pi_x(z)$: Proximity of an instance z from x .

$\Omega(g)$: A measure of complexity of the explanation $g \in G$.

The goal is to minimize the locality aware loss L without making any assumptions about f , since a key property of LIME is that it is model agnostic. L is the measure of how unfaithful g is in approximating f in the locality defined by π_x .

4. Empirical results

The following sections including 6 parts: the description of ICFS and validity test, division of financial risk status, evaluation of the early warning model, the feature importance of the early warning model, global interpretation of the early warning model, and local interpretation of the early-warning model.

4.1. The description of ICFS and validity test

The description of ICFS and validity test is for describing ICFS and compare the specific values of ICFS with real risk events in history to verify the plausibility of ICFS.

This paper obtained the ICFS based on daily frequency data, some extreme value may bring bad effect, so smooth ICFS trend items (Fig. 2) are obtained by HP filtering (Hodrick & Prescott, 1997). The dark gray fluctuation area for ICFS chart, black curve for HP filter ICFS trend

items, light gray rectangular shadow part represents the warning period of financial stability (the high-risk status is divided by Markov regime switching model), which indicating large financial pressure and beyond the safety range.

Table 2 is a summary of the financial stability early-warning period. It contains the start point of each warning period, the extreme index interval and the risk events in the current period. In Table 2, the warning period lasted from December 10, 2007 to January 21, 2009 was the longest, and the extreme difference of the ICFS reached 0.5527, which indicates that during U.S. subprime mortgage crisis, the ICFS was greater than that in other periods, reflecting the severe impact of the subprime mortgage crisis on various financial markets in China.

This table presents the six time periods in high-risk status (identified by Markov regime switching model), the ICFS interval, range, corresponding risk events.

4.2. Division of financial risk status

Division of financial risk status is for dividing ICFS into three categorical variables including low risk, medium risk and high risk in order to use the categorical variables as output labels of the XGBoost model.

In the final estimation results by Markov Regime Switching model, the mean values of ICFS in low, medium and high risk status are 0.0520, 0.1101, 0.2797, while the variances are 0.0003, 0.0015, and 0.0093. It can be seen that the mean and variance of the ICFS in high risk status are high, indicating that the financial risk in this period is relatively high and deviating from the norm, it is more prone to fluctuations.

Table 2
Financial stability early warning range and its reasons.

Time starting point	Index interval	Range	Risk events
2007.12.10–2009.1.21	[0.0420,0.5947]	0.5527	The U.S. subprime mortgage crisis in 2008
2011.7.26–2012.4.24	[0.0760,0.4758]	0.3998	The European debt crisis in 2012
2013.6.14–2013.8.29	[0.1036,0.3692]	0.2656	The first money shortage in China's banking industry in 2013
2013.12.31–2014.2.7	[0.1038,0.2600]	0.1562	The second money shortage in China's banking industry in 2013
2015.12.29–2016.1.29	[0.1286,0.3326]	0.2041	Stock market crash in 2015–2016
2018.6.19–2018.10.29	[0.0448,0.4154]	0.3706	China-US trade friction in 2018

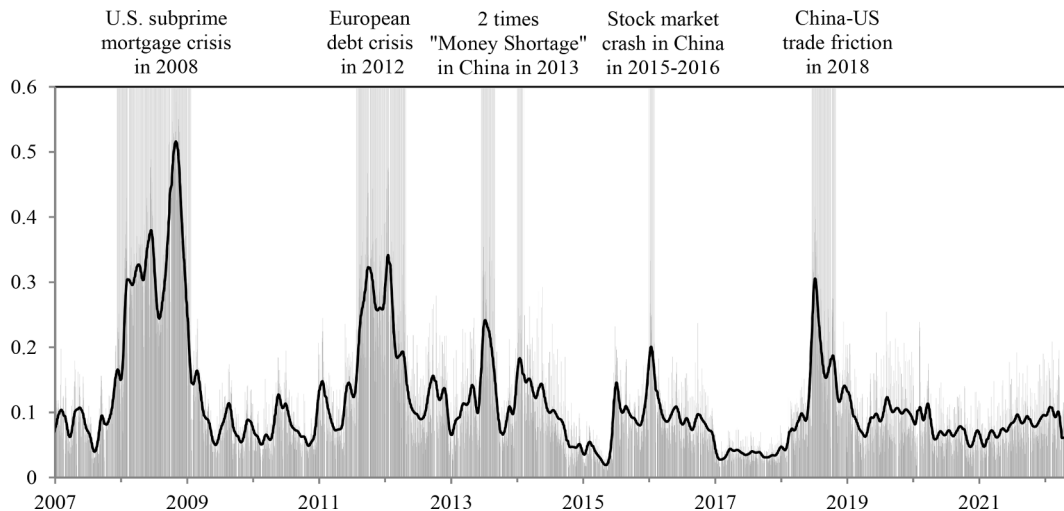


Fig. 2. Trend of ICFS and warning range.

The regime switching probability matrix output by the model is shown in Table 3, (i) The three regimes have higher probabilities in their own status (0.8824, 0.9057, 0.9653), which indicating that financial risks will remain stable once they enter a certain status, and it is not easy to change to other status in a short time. (ii) It can be seen that the probability of mutual transfer between low and medium risk regimes (0.0835, 0.1170) is higher than the probability of mutual transfer between medium and high risk regimes (0.0347, 0.0108), which indicates that low and medium risks are relatively common status, and high-risk periods occur relatively rarely, but once entering the high-risk regime, it is difficult to return to normal in a short period of time. (iii) The probability of mutual transfer between low and high risks (0.0000, 0.0006) is less than 1%, indicating that it is almost impossible to switch between low and high risk status, but it is relatively easier to transfer to adjacent status. In order to facilitate the analysis, this paper combines the smoothed probabilities of three kinds of status into one figure, as shown in Fig. 3, where the smoothed probability represents the transition probability of two consecutive periods being the same status.

Fig. 3 shows that the smoothing probability of low, medium, and high fluctuations is reflected in order from top to bottom. According to the smoothing probability of different fluctuations, this paper divides ICFS into three categorical variables: low risk, medium risk and high risk. In general, (i) the frequency of medium risk in the samples is higher. (ii) Based on the low-fluctuation smoothing probability diagram, it can be found that from October 2004 to June 2005, throughout the entire year of 2007, and at the end of 2020 and early 2021, ICFS remained the status of low volatility. It shows that preventing and defusing financial risks have achieved good results during these periods. (iii) Based on the high-fluctuation smoothing probability diagram, Fig. 3 shows that from 2008 to early 2009, the second half of 2011 to the first half of 2012, mid and late 2013, early 2016 and late 2018 were all classified as high-risk status, this is almost completely consistent with the analysis of the validity test mentioned in previous part.

4.3. Evaluation of the early warning model

Evaluation of the early warning model is for evaluating the performance of XGBoost model, which also tests if XGBoost model has a good early warning effect.

In this paper, we build an early warning model with 20 basic indicators in the money market, stock market, bond market and foreign exchange market as the input features, and the three categorical status of low, medium and high financial risk (0, 1 and 2) as the output labels.

Firstly, samples were divided into the training set and the test set in a ratio of 7 to 3, and the early warning effect was optimized by adjusting the parameters. The accuracy, precision, recall and F1 score can be calculated from the prediction results of the test set.

Confusion matrix is a visual tool to evaluate the performance of classification models. Each column of the confusion matrix represents the prediction values of the model, each row represents the true values of the samples, and the positive diagonal represents the number of correctly classified samples. The confusion matrix obtained from the XGBoost model is shown in Fig. 4.

According to Fig. 4, there are 199 high-risk samples, 758 medium-risk samples, and 151 low-risk samples were correctly classified among the 1,170 samples in the test set. The proportion of correctly

Table 3
Regime switching probability matrix.

	Low risk regime	Medium risk regime	High risk regime
Low risk regime	0.8824	0.0835	0.0000
Medium risk regime	0.1170	0.9057	0.0347
High risk regime	0.0006	0.0108	0.9653

This table presents the Markov regime switching probability matrix, it indicates the probability of transferring from one status to another status.

classified samples was 94.7%, and only a small number of samples were still misclassified into adjacent categories. The number of samples misclassified as high, medium and low risk status is 11, 43 and 8, and the total number of misclassified samples is only 5.3% of the total samples. Therefore, we show that the XGBoost model has a good performance in early warning effect in financial stability.

In addition, from the evaluation of the accuracy, macro accuracy, macro recall and macro F1 of the early warning model, it reached 0.9470, 0.9479, 0.9171, 0.9316, which also shows that the XGBoost model has a good early warning effect.

In order to enhance the interpretability of the early warning model, interpretable tools such as feature importance, partial dependency plots, SHAP and LIME are introduced, aiming at building a comprehensive interpretation framework containing global and local interpretation methods to explain the intrinsic mechanism of the “black-box model”.

4.4. The feature importance of the early warning model

The feature importance of the early warning model is for demonstrating the feature contribution of different variables in XGBoost model to find out the most important features.

With the continuous development of machine learning algorithms, more and more integrated algorithms gradually improve the prediction accuracy of early warning models. However, the limited interpretability of deep learning models may create “black-boxes,” whose opaque inner-workings mask how inputs relate to outputs. It is obvious that the more complex the model is, the higher the “black box” degree of machine learning, which means it is difficult to interpret those complex models. To compensate for the lack of interpretability of machine learning, this paper uses the SHAP (SHapley Additive exPlanations) interpretable tool introduced by Lundberg and Lee (2017) to explain the early warning model. SHAP summary plot is an effective tool to describe the overall relationship between financial risks and factors. The role of SHAP is to measure the marginal contribution of the input features in machine learning model.

Taking the training set samples as an example, the input features are the 20 basic indicators in 6 sub-markets, and the output labels are the 3 categorical variables (low, medium and high risk status). According to the different output labels, the SHAP summary plot can be divided into three kinds. We usually only consider the more practical SHAP summary plot that describes high-risk status. In Fig. 5, if the SHAP value corresponding to the value of the feature j of the sample i is positive, it means that the marginal contribution of the feature j is positive to the warning of sample i as a high-risk status, in other words, when the feature j of sample i takes this positive value, sample i is more likely to be predicted as a high-risk status. The SHAP summary plot of the high-risk status presented in this paper is shown in Fig. 5.

The role of the SHAP summary plot is to rank the feature contribution of the warning model from a global perspective. In Fig. 5, it shows the names of the features on the left, and the right color is from blue to purple to red, indicating the value of the feature from low to high. The horizontal axis represents the SHAP values, the value range and size of the SHAP values corresponding to the individual features for each sample. Each row in Fig. 5 represents one feature, each point represents one sample, number of sample points on each row are the same (each row contains 2,728 samples), and wide areas represent more sample points than narrow areas. Take the first row of Fig. 5 as an example, INDI_CMAX is ranked first, indicating that INDI_CMAX has the highest contribution to the early warning model. Further, when INDI_CMAX value is small (blue area), the SHAP value is negative, indicating that the sample point with small value is more likely to be predicted to high risk status, and the SHAP value gradually becomes positive as it increases (purple and red area), indicating that the sample point is predicted to be high risk gradually increases when INDI_CMAX value is large.

Besides, 180FINI_CMAX, credit_spread, REI_CMAX, term_spread, 7D_reporate, COMI_CMAX and REI_illiquidity also have similar

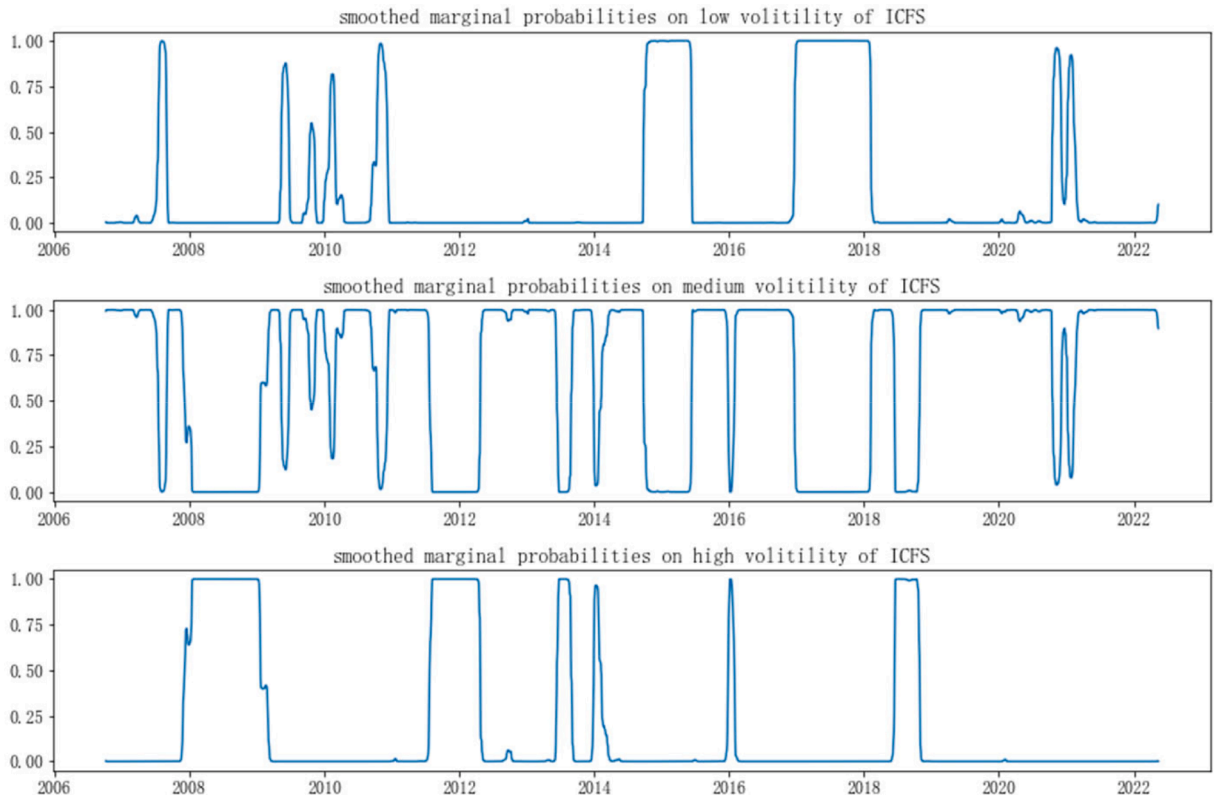


Fig. 3. Probabilistic diagram of the Markov regime switching model.

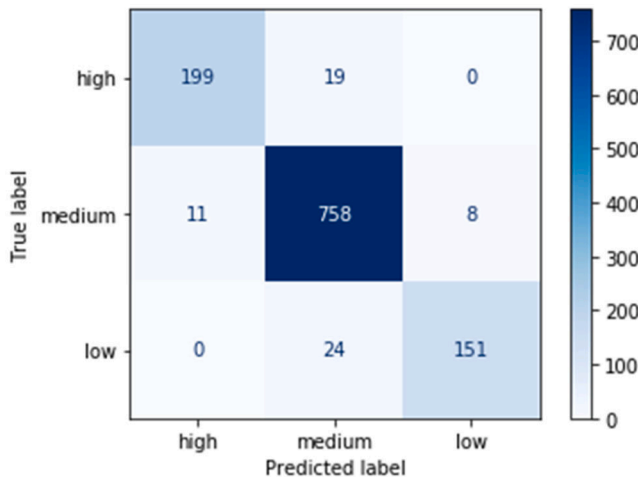


Fig. 4. The confusion matrix of XGBoost.

characteristics, blue left purple medium and red right is presented on the image. The deeper the red color is, the larger value of indicator is, and the positive SHAP value can explain the prediction of high risk status. In other words, the larger the value, the higher the probability that the sample is predicted to be at high risk.

4.5. Global interpretation of the early warning model

Global interpretation of the early warning model is for exploring how the features affect the output of the model, and showing the non-linear relationship among the variables.

In order to more intuitively explore how the features affect the output of the model, and excavate valuable information to assist the regulatory monetary authorities to take targeted measures, this paper

uses the SHAP value mapping plots to show the non-linear relationship among the variables. Unlike the partial dependency plots, the ordinate of the SHAP value mapping plots is SHAP value but not the output label value. The SHAP values represent the marginal contribution of a feature to output label, in Fig. 6 we give the change in the probability with a feature changes, and the SHAP value mapping plots for the first 8 significant features extracted by the XGBoost model are enumerated.

As shown in Fig. 6, the left vertical axis and the horizontal axis are the SHAP value and the range of values for that single feature. From the general trend, the eight important features increased as the value increased, the SHAP value increased, and the probability of the sample being predicted to be in high risk status also gradually increased. A horizontal line made from the value of SHAP is 0 will pass through the two points on the graph, we call them safe point and warning point (safe point and warning point may be the same point), and two vertical lines made from these two points will divide the whole graph into three intervals, low risk, medium risk and high risk. In existing articles SHAP mapping plots are used for analyzing the effect trend of a feature on SHAP value, but in this paper we innovatively point out the warning point and safe point of the feature, which can help us to forecast financial stability. As shown below, we explain in detail the SHAP mapping plots of the most important 8 features.

4.5.1. Mapping relationship between *INDI_CMAX* and high risk status

In Fig. 6a, (i) When the value of *INDI_CMAX* is in the interval of $[0, 0.145]$, it means that the SSE industrial index is down less than 14.5% from its peak in the past year, this shows that the market risk in the industrial sector is relatively small, and negative SHAP value means that the probability of the sample being predicted to be in high risk status will decrease. (ii) When the value of *INDI_CMAX* is in the interval $[0.145, 0.75]$, it means that the SSE industrial index fell more than 14.5% from its peak in the past year, and positive SHAP value reflects positive impact on the early warning model. It can be seen that the greater the value of *INDI_CMAX*, the more that the sample will be likely

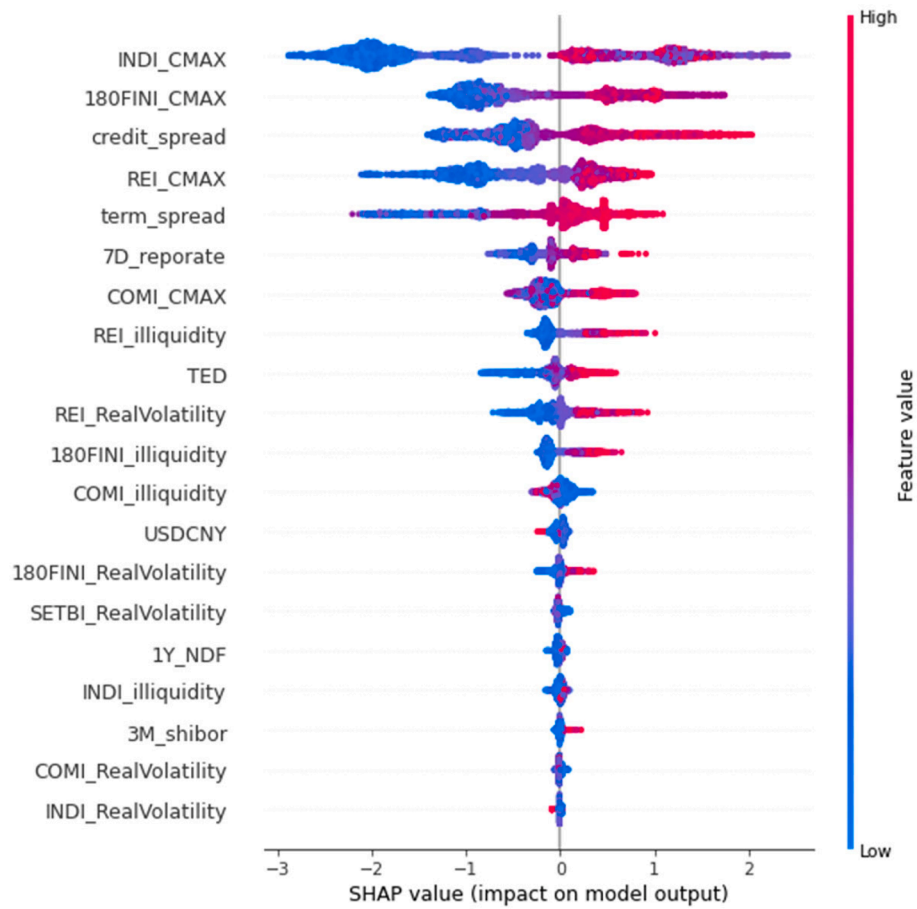


Fig. 5. A SHAP summary plot of the high risk status.

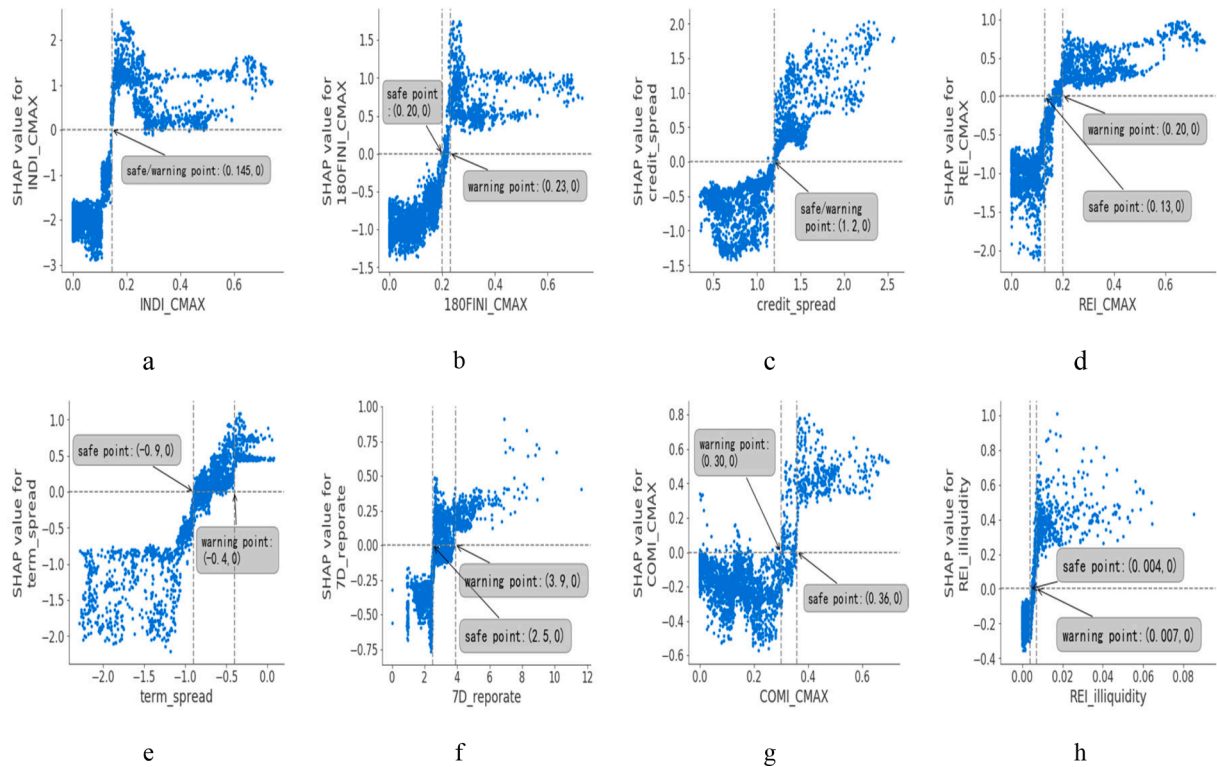


Fig. 6. SHAP value mapping plots for the top 8 important features based on the XGBoost model.

to be predicted in high-risk status. This also shows that risks from the real economy may be the biggest source of financial risk in China, which is consistent with Financial Stability Bureau of the People's Bank of China pointed out that "many financial risks come from the real economy".

4.5.2. Mapping relationship between 180FINI_CMAX and high risk status

In Fig. 6b, (i) When the value of 180FINI_CMAX is in the interval of [0, 0.20], it means that the 180 financial index is down less than 20% of its peak in the past year, and negative SHAP value indicates that the market risk in the financial sector is relatively small, and the sample is less likely to be predicted to be in high risk status. (ii) When the value of 180FINI_CMAX is in the interval of [0.23, 0.75], it means that the 180 financial index is down more than 23% from its peak in the past year, and the positive SHAP value indicates the high probability that the samples will be more likely to be predicted in high risk status. It can be seen that the greater the 180FINI_CMAX, the greater the probability that the sample will be divided into high risk status. This shows that ensuring the healthy operation of banks, securities companies, insurance companies and other financial institutions, and maintaining the overall stable and healthy development of the financial industry, will play a positive role in maintaining financial stability.

4.5.3. Mapping relationship between credit_spread and high risk status

In Fig. 6c, (i) When the value of credit_spread is in the interval of [0.25, 1.2], it means that the yield on AAA corporate bonds is higher than the treasury bond within 1.2%, the negative SHAP value indicates that the credit_spread reduces the probability that the sample is predicted to be in high risk status and credit_spread runs well in this interval. (ii) When the value of credit_spread is in the interval of [1.2, 2.5], it means that AAA corporate bonds yield is higher than national bonds over 1.2%, and the positive SHAP value indicates that the sample is more likely to be predicted to be in high risk status. The main reason is that most investors lack confidence in corporate bonds during the economic recession period, and they are more willing to invest in high-credit treasury bonds to avoid risks. In addition, enterprises need to raise interest rates in order to successfully issue bonds, thus generating high credit spreads. It can be seen that the high credit spread is the performance of the real economy depression and the lack of market expectations.

4.5.4. Mapping relationship between REI_CMAX and high risk status

In Fig. 6d, (i) When the value of REI_CMAX is in the interval of [0, 0.13], it means that the real estate fell less than 13% from its peak in the past year, and the negative SHAP value indicates that there is less risk and more financial stability in the real estate market. (ii) When the value of REI_CMAX is in the interval of [0.2, 0.75], it means that real estate index fell more than 20% from its peak in the past year, and the positive SHAP value shows that the real estate market is too risky. This also indicates that stabilizing housing prices, supporting residents' reasonable housing demand and maintaining the steady and healthy development of the real estate market is of great significance to preventing financial risks.

4.5.5. Mapping relationship between term_spread and high risk status

In Fig. 6e, (i) when the value of term_spread is in the interval of [-2.25, -0.9], it means that difference between long-term and short-term Treasury bond yields is more than 0.9, and the negative SHAP value indicates the low risk. (ii) when the value of term_spread is in the interval of [-0.4, 0], it means that difference between the yield of long and short term bonds is less than 0.4, and the positive SHAP value indicates a greater risk. This suggests that the smaller the yield difference between long-term and short-term Treasury bond, the more pessimistic market expectations, and the greater the financial risk. This further shows that stabilizing expectations is of great significance to preventing financial risks.

4.5.6. Mapping relationship between 7D_reporate and high risk status

In Fig. 6f, (i) when the value of 7D_reporate is in the interval of [0, 2.5], it means that the 7-days repo rate in the interbank market is below 2.5%, the banking liquidity is good, and the negative SHAP value indicates that the index runs well in this range with low financial risk. (ii) when the value of 7D_reporate is in the interval of [3.9, 12], it means that the 7-days repo rate in the interbank market is above 3.9%, indicating that the liquidity risk in the short-term financing market increases, and the positive SHAP value indicates that the probability of the sample is more likely to be predicted in high risk status. Therefore, maintaining the reasonable liquidity of the banking industry is of positive significance to prevent financial risks.

4.5.7. Mapping relationship between COMI_CMAX and high risk status

In Fig. 6g, (i) when the value of COMI_CMAX is in the interval of [0, 0.30], it means that the decline of the SSE commercial index is less than 30% compared with the peak in the past year, and the negative SHAP value indicates that the risk of the commercial sector is low. (ii) when the value of COMI_CMAX is in the interval of [0.36, 0.75], it means that SSE commercial index fell more than 36% compared with the peak in the past year, and the positive SHAP value indicates that the risk of the commercial sector is increasing. This further demonstrates that maintaining the healthy operation of the real economy sector plays an important role in preventing financial risks.

4.5.8. Mapping relationship between REI_illiquidity and high risk status

In Fig. 6h, (i) When the value of REI_illiquidity is in the interval of [0, 0.004], it means that the yields on real estate stocks are low, the housing bubble is smaller, and the negative SHAP value indicates that the financial risk is relatively small. (ii) When the value of REI_illiquidity is in the interval of [0.007, 0.08], it means that the yields of real estate stocks is high. As we all know, the higher the yield, the more likely lots of capital will enter the property market to speculate, it will cause an asset mismatch and increasing of housing bubble, the positive SHAP value indicates that the financial risk is correspondingly large. This also shows that it is of great significance to resolutely implement the "housing without speculation" and continuously curb the real estate financial bubble to prevent financial risks.

In summary, the points where SHAP value is 0 corresponding to the safe point and warning point of the warning model, the left area on safe point is the low risk range, the SHAP value between the safety point and the warning point is positive but the value is small, which can be considered as the medium risk range, and once the value of a feature exceeds the warning reference value, the SHAP value will increase significantly, resulting in the financial risk increase. In summary, the warning reference values based on the SHAP value mapping plots are shown in Table 4.

This table presents the 8 most important variables in the XGBoost model, and lists their feature value interval and the early warning reference values respectively by SHAP value mapping plots.

4.6. Local interpretation of the early-warning model

Local interpretation of the early-warning model is for analysis of

Table 4

Early warning reference values based on the SHAP value mapping plots.

Feature name	Interval of feature value	Early warning reference value
INDI_CMAX	[0, 0.75]	0.145
180FINI_CMAX	[0, 0.75]	0.23
credit_spread	[0.25, 2.5]	1.2
REI_CMAX	[0, 0.75]	0.2
term_spread	[-2.25, 0]	-0.4
7D_reporate	[0, 12]	3.9
COMI_CMAX	[0, 0.75]	0.36
REI_illiquidity	[0, 0.08]	0.007

single sample, in this part we can identify the marginal contribution of each features and explain the effect is positive or negative.

In this paper, SHAP force plots are used to locally explain a single sample, and the 40th sample in the training set (564 in the original data set) is randomly selected. As shown in Fig. 7, the single sample is predicted to be in low, medium and high risk status in the SHAP force plots from top to bottom. (i) The top horizontal axis coordinate indicates the possible value interval of the SHAP values of all the samples, the base value is the mean of the SHAP values for all samples. (ii) The value after the feature name is the actual feature value of the sample. (iii) The red and blue area indicates the direction and size of the SHAP value of a sample's different feature, red (blue) color corresponds to the positive (negative) SHAP value, wide (narrow) corresponds to large (small) of SHAP value. (iv) Numbers (-2.13, -0.60 and 3.15) adjacent to the red and blue region indicate the SHAP output values when the sample is predicted to be in a low, medium, and high risk status.

In this paper, softmax objective function is used in the XGBoost model to deal with multi-classification problems, and the softmax function is a generalization of the binary classification function sigmoid on multi-classification, on purpose to show the results of multi-classification in the form of probability. The early warning results of low, medium and high risk are converted into probability. Firstly, use softmax function to convert the predicted value into an exponential form with base e , $y_1 = e^{x_1} = e^{(-2.13)} = 0.1188$, $y_2 = e^{x_2} = e^{(-0.60)} = 0.5488$, $y_3 = e^{x_3} = e^{3.15} = 23.3361$, and then normalize the results to probabilities,-

$P_1 = y_1 / \sum_{i=1}^3 y_i \approx 0.0049$, $P_2 = y_2 / \sum_{i=1}^3 y_i = 0.0229$, $P_3 = y_3 / \sum_{i=1}^3 y_i = 0.9722$, the results show that the probability of this random sample being predicted as low risk is 0.49%, 2.29% for medium risk, and 97.22% for high risk, the probability of being predicted as high risk sample is largest, while its true value as the label showed is 2 (the high risk sample).

Besides, as can be seen from the SHAP force plot of the high risk status (the picture at the bottom) in Fig. 7, (i) INDI_CMAX is the feature that makes the greatest marginal contribution to the early warning results, followed by term_spread, REI_CMAX, 180FINI_CMAX and credit_spread, all the features jointly determine the risk status result of the sample. (ii) The red features indicate that the effect on SHAP value is positive, blue features effect on SHAP value is negative, so on the one hand it is obvious that REI_CMAX, 180FINI_CMAX, credit_spread and COMI_CMAX have positive effects on SHAP value, the higher the values of these five feature variables are, the more the probability that the sample is predicted as high risk will be increased. On the other hand, term_spread and TED have negative effects on the SHAP value, so the higher the feature values of term_spread and TED are, the lower the

financial risk of this sample will be.

5. Robustness test

This study test if the empirical conclusion is robust by several methods, including adjustment of the parameters in XGBoost model, replacing other machine learning models such as the decision tree, support vector machine, random forest and artificial neural network, replacing the sorting method of feature importance, replacing global interpretation methods and local interpretation methods of early warning model.

5.1. Adjustment of XGBoost parameter

In this paper, the optimal parameters were determined by grid search and cross-validation, specific parameters that can improve the model accuracy are as follows: (i) learning rate, used to prevent overfitting, the weight of the new feature is obtained directly after each boosting calculation, learning rate makes the boosting calculation process more conservative by reducing the weight of features; (ii) gamma, used to specify the minimum loss function drop value required for node splitting, when a node divides, only the value of the loss function drops after division, the node will be divided; (iii) max_depth, used to avoid overfitting, the larger the max_depth is, the model will learn more specific and local samples; (iv) subsample, used to control the proportion of random sampling for each tree, lower the parameter values will lead a more conservative algorithm to avoid overfitting. (v) colsample_bytree, used to control the number of columns sampled by each tree (each column is a feature); (vi) reg_lambda, the L2 regular term parameter used to control the weight value of the model complexity, the larger the parameter is, the less likely the model will overfitting.

Accuracy of XGBoost model in the previous paper has reached 94.70%, to verify whether the setting of the parameter range affects the accuracy, the learning rate is modified (learning_rate drops from 0.3 to 0.1), here we redefine the optimal parameters based on the new range. The results are shown in Table 5, the accuracy of the model reached 94.53%, which decreased by 0.17% compared to the result before the adjustment. The accuracy, macro precision, macro recall, and macro F1 were 0.9450, 0.9545, 0.9063, and 0.9285. Under the new parameters, the accuracy of the model can be basically consistent with the previous result, which shows that the early warning result is robust.

This table presents the parameters that need to be optimized in XGBoost model, range of values for parameter optimization, and the optimal parameters.



Fig. 7. SHAP force plots of one single sample (low, medium and high risk are shown from top to bottom).

Table 5

Optimum parameters if changing the learning rate from 0.3 to 0.1.

Parameter name	Parameter range	Optimal parameters
n_estimators	0–1000	49
max_depth	3–10	5
min_child_weight	1–6	1
gamma	(0,0.1,0.2,0.3,0.4,0.5)	0.1
subsample	(0.6,0.7,0.8,0.9,1)	0.8
colsample_bytree	(0.6,0.7,0.8,0.9,1)	0.7
reg_lambda	(0.00001,0.0001,0.001,0.01,0.1,1,10,100)	0.001

5.2. Other machine learning models

In order to verify if the warning performance of XGBoost model is robust, this paper introduces four other models including Decision Tree (DT), Support Vector Machine(SVM), Random Forest(RF), Artificial Neural Network(ANN) as comparison. For each model, we tested the warning performance of the above features on financial stability, and verify if XGBoost model has better performance.

Table 6 shows the assessment of the early warning effect under the different models, compared with the XGBoost model, in accuracy, macro precision, macro recall, and macro F1. DT was reduced by 3.16%, 5.44%, 2.51%, 3.89%; SVM was reduced by 3.25%, 2.24%, 5.79%, 4.3%; RF was reduced by 8.29%, 1.89%, 17.99%, 12.91%; ANN was reduced by 0.94%, 1.75%, 1.03%, 1.36% respectively.

To sum up, XGBoost outperformed the comparison model in each measure metrics, which can prove that the predicted effect of these several models is ranked as XGBoost > ANN > DT > SVM > RF. In addition, it can be found that the ensemble classifier (XGBoost, RF) performs better than single classifier (DT, SVM). This is because the prediction results of the ensemble classifiers are usually formed by a number of weak classifiers in parallel or iteratively, which are formed by a linear combination of single classifiers, and the wrong prediction results are constantly corrected to achieve better results.

5.3. Feature importance ranking based on XGBoost

Feature importance could further explain which features make more contributions to the prediction results under the XGBoost model. As it shown in Table 7, INDI_CMAX has the highest relevance to risk status and its feature importance reached 14.59%. In addition, there are 2 more features' feature importance exceeds 10%, including term_spread and 180FINI_CMAX. The feature importance ranking is shown in Table 7. The feature importance of the first 8 features can reach 72.64% in all, which can explain the prediction results to a large extent, indicating that these features are the main influencing factors of financial stability. Compared with the SHAP summary plot, the first nine features are ranked exactly the same except for the term_spread, which indicates that the feature importance ranking output by the XGBoost model has robustness.

This table presents the feature importance rank of 20 variables in XGBoost model, the sum of all the feature importance values equal 1.

Table 6

Evaluation of early warning effect based on different models.

Model	Accuracy	Macro precision	Macro recall	Macro F1
XGBoost	0.9470	0.9479	0.9171	0.9316
DT	0.9154	0.8935	0.8920	0.8927
SVM	0.9145	0.9255	0.8592	0.8886
RF	0.8641	0.9290	0.7372	0.8025
ANN	0.9376	0.9304	0.9068	0.9180

This table presents the performance of various evaluation metrics, XGBoost is the main model, the 4 other machine learning models are for comparison.

5.4. Global interpretation based on the partial dependency plots

The Partial Dependency Plots (PDP) show the non-linear relationship between the feature variables (x) and the amount of change (y) of the predicted value, reflecting the marginal contribution of a certain feature to the prediction result. Fig. 8 shows the deviation degree of the predicted value from the reference value. If the predicted value increases as one feature increases, so the feature contributes positively the prediction result, otherwise, the feature plays a negative role in the prediction result. To verify the robustness of the results in the above SHAP value mapping plots, the partial dependence plots of the first 8 important features are also displayed in Fig. 8.

As shown in Fig. 8, the y-axis represents the amount of change in the predicted value of the label compared to the benchmark value, and the shaded part represents the confidence interval. From the overall trend, the probability that the sample is predicted to be in high risk status increases as the feature values increase. Taking Fig. 8a as an example, it can be seen that when the value of INDI_CMAX is in the range of [0, 0.12], the financial risk is extremely low; but in the range of [0.12, 0.17], the financial risk increases significantly. When the value is greater than 0.17, the financial risk will always maintain in the high risk status. Similarly, the influence trend of other important features on the predicted probability is roughly similar. As for term_spread, the risk rises rapidly at a value greater than -0.7, and always maintains at the high risk status at the value of -0.4. The probability trends of other variables will not be described here because of the limitation of space. In addition, if the high risk probability in the graph is raised to a highest position and no longer changes dramatically, the corresponding feature value here can be regarded as the warning reference value of the variable in financial stability period. Once the sample exceeds this reference value, it will always remain in the high risk status. According to Fig. 8, the reference warning value of each feature is roughly as follows, INDI_CMAX = 0.17, 180FINI_CMAX = 0.24, credit_spread = 1.3, REI_CMAX = 0.21, term_spread = -0.4, 7D_reporate = 4, COMI_CMAX = 0.38, REI_illiquidity = 0.01. It is almost the same as the warning value obtained from the previous SHAP value mapping plots (as shown in Table 8), which can indicate that the warning reference value obtained based on the XGBoost model is robust.

This table presents the early warning reference value in SHAP value mapping plots (already showed in Table 4) and in PDP, a comparison between them shows that the difference in values is not significant.

5.5. Local interpretation based on LIME

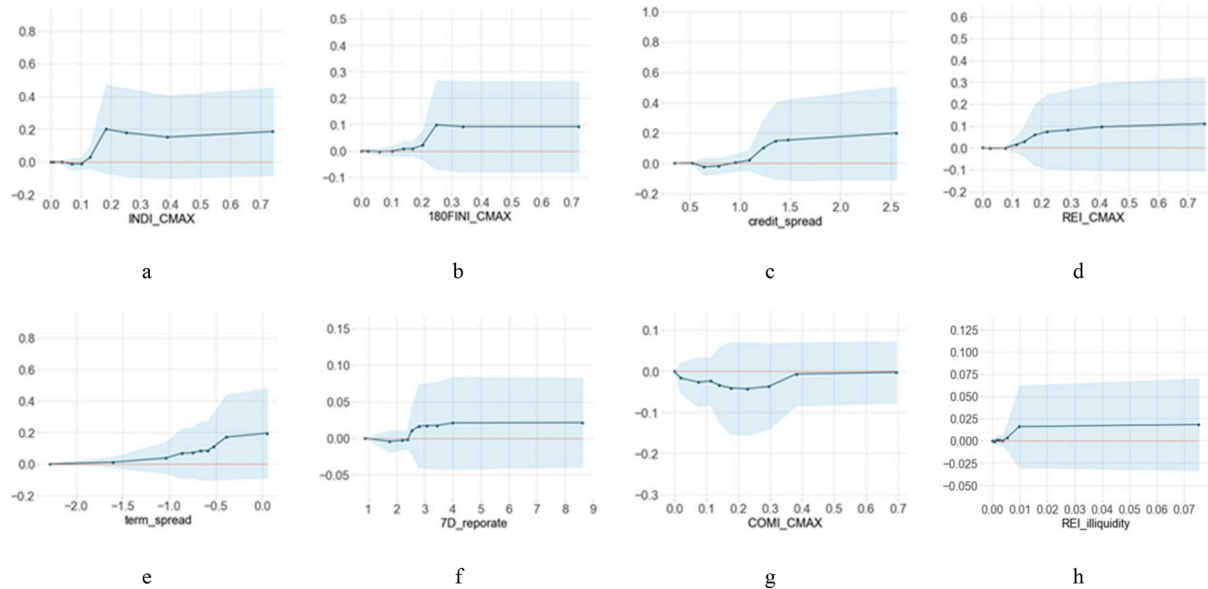
Here we use the same sample selected in the SHAP force plot analysis mentioned above, the 40th sample in the training set (the index was 564 in the original data set), and the visual interpretation of the predicted value based on LIME is shown in Fig. 9.

In Fig. 9, it can be found (i) the prediction probabilities on the left indicate that the probabilities of the sample being predicted as low, medium and high risk are 0%, 2% and 97%, which is almost the same as the results obtained in the SHAP value mapping plot (0.49%, 2.29%, and 97.22%). (ii) The tree graph on the right reflects the ranking of the contribution values of the six most important features, where green parts represent the variables that would promote samples to be divided into high risk status, blue parts indicate the variables that would restraint samples to be divided into high risk status. It is obvious that the outcomes are consistent with the results obtained by the SHAP value mapping plots. What's more, the values outside the rectangle represent the contribution of each feature, contribution values from top to bottom in the tree graph are 0.16, 0.15, -0.14, 0.11, -0.04, 0.04. The contribution values of these features are important bases for supporting or against the sample being divided into high risk status, and the features largely overlapping with the features exhibit in the SHAP force plots. Overall, the local interpretation results based on the XGboost model have robustness.

Table 7

A ranking table of feature importance based on the XGBoost model.

Variable	Feature importance	Rank	variable	Feature importance	Rank
INDI_CMAX	0.1459	1	1Y_NDF	0.0300	11
term_spread	0.1346	2	3M_shibor	0.0293	12
180FINI_CMAX	0.1288	3	180FINI_illiquidity	0.0291	13
credit_spread	0.0875	4	REI_illiquidity	0.0280	14
REI_CMAX	0.0783	5	UCDCNY	0.0231	15
7D_reporate	0.0572	6	COMI_illiquidity	0.0166	16
COMI_CMAX	0.0561	7	180FINI_RealVolatility	0.0133	17
TED	0.0370	8	SETBI_RealVolatility	0.0123	18
REI_illiquidity	0.0366	9	INDI_RealVolatility	0.0122	19
INDI_illiquidity	0.0330	10	COMI_RealVolatility	0.0110	20

**Fig. 8.** Partial dependency plots of the first 8 important features based on the XGBoost model.**Table 8**

Comparison of early warning reference values based on the SHAP value mapping plots and PDP.

Variable name	Early warning reference value in SHAP value mapping plots	Early warning reference value in PDP
INDI_CMAX	0.145	0.17
180FINI_CMAX	0.23	0.24
credit_spread	1.2	1.3
REI_CMAX	0.2	0.21
term_spread	-0.4	-0.4
7D_reporate	3.9	4.0
COMI_CMAX	0.36	0.38
REI_illiquidity	0.007	0.01

6. Conclusions and implications

6.1. Conclusions

In this paper, we firstly construct ICFS based on the data of 20 indicators from the money market, stock market, bond market and foreign exchange market. Then ICFS is divided into three risk status (low, medium and high) by Markov regime switching model. What's more, construction of early warning model is done through XGBoost, and 8 main influencing factors are extracted through SHAP interpretation method. Finally, in our study we introduce several approaches to test if the results are robust in the end.

According to the empirical results, a few conclusions are drawn.

Firstly, we construct ICFS by dynamic weighting method based on the time-varying correlation coefficient, and ICFS in this paper is consistent with the actual situation, can accurately highlight the major risk events that have occurred in the history, such as U.S. subprime mortgage crisis in 2008, European debt crisis in 2012 and so on. Secondly, we also find that compared with other machine learning models such as SVM, decision tree, random forest and ANN, the prediction performance of the XGBoost model is significantly better than simple weak classifiers, demonstrates higher accuracy and stronger generalization ability. It shows that the XGBoost model has a very good applicability in early warning in China's financial stability. Finally, we find that assigning importance to the individual variables is a challenging task because it depends on the selection of models and interpretable tools. Based on XGBoost and through the Visualization analysis of results of series interpretable tools such as SHAP, PDP and LIME, we found that the 8 most important factors affecting financial stability are INDI_CMAX, 180FINI_CMAX, credit_spread, REI_CMAX, term_spread, 7D_reporate, COMI_CMAX, REI_illiquidity. These variables all have a positive impact on financial risk, in other words, the larger the value of these variables, the more they can promote the sample points to be classified as high risk ranges, which means that the degree of financial stability is lower. It indicates that the downturn in real economy, the unstable operation of financial institutions, insufficient market expectations and the bubble of the real estate market are the main risk factors affecting China's financial stability.

To conclude, this paper extends the financial stability literature by serving a machine learning early warning model by using daily high frequency data, and demonstrated the effectiveness of multiple

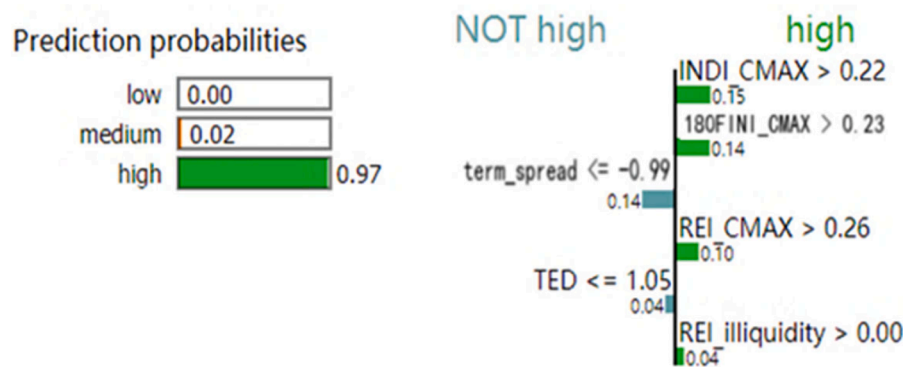


Fig. 9. Prediction probability plots based on single sample by LIME.

interpretable tools. They are such universal tools that could be well applied to the research process of financial stability in other countries. For example, an index of financial stress for Canada developed by Illing and Liu (2006), the financial conditions indices for G7 countries proposed by Goodhart and Hofmann (2001), the financial stability index for Colombia constructed by Morales and Estrada (2010), the financial condition indexes for the United States and euro area created by Matheson (2012) all could be put into the machine learning model and interpretable framework to get further research. In addition, the research idea of this paper has certain practical significance that it can also be applied to stock market forecasting, fraud prediction, bank customer relationship management, etc. Therefore, we hope the application of the interpretable framework including SHAP, PDP and LIME in this paper, as well as the robustness test on the effect of the model in section 5 could be analytical reference provided for machine learning investigations. What's more, future research can focus more on the explainable analysis of machine learning, so that machine learning can ensure accuracy while gradually showing its internal mechanism, which can be well applied to various fields of real life.

Nevertheless, this study contains several limitations. Firstly, from the point of view of the construction of the index, we only considered financial risks in periods of falling stock indices but neglected the financial risk during stock indices rising. Secondly, this paper may not be sufficient in the selection of indicators, and further research can add variables related to government policy implementation on the current basis to measure the impact of policy implementation on the financial stability. Therefore, future attempts should be made to fill the gap.

6.2. Implications

The conclusions of this paper can bring the following implications for government and financial supervision departments.

Firstly, it has been confirmed that real economy performance indicators such as INDI_CMAX and COMI_CMAX in this paper will significantly influence financial stability status. From the real economy affect the financial stability transmission path, financial risk generally due to enterprises structural excess capacity, excessive leverage and asset mismatch, leading the financial risk speed up the transmission and agglomeration to the financial sector. Therefore, the government and financial supervision departments need to resolve the overcapacity in traditional industries, expand effective demand based on domestic demand, improve the financial service system that supports the development of the real economy, strengthen the support of finance and taxation to entities, and create a good production and operation environment. Support the healthy development of the real economy and resolve systemic financial risks with high-quality economic development.

Secondly, the indicators of the operation of financial institutions such as 180FINI_CMAX, 7D_reporate in this paper play important roles in financial stability. From the perspective of financial institutions' risk

sources, it mainly from high leverage, maturity mismatch, excessive innovation in financial market, lake of liquidity, financial bubble inflation, etc. Due to the particularity of the financial industry, once a risk event occurs, it will spread within the financial industry and overflow to other sectors, and the risk of individual financial institution may evolve into systemic financial risk. Therefore, the government and financial supervision departments need to reverse the trend of financial assets deviating from real to virtual, prevent the contagious spillover of various financial risk points, consolidate the long-term mechanism to prevent and defuse financial risks, and improve the transparency of financial supervision and the level of the rule of law. Through these measures to guide finance to truly return to the origin of serving the real economy, and maintain the overall stable and healthy development of the financial industry.

Thirdly, indicators such as credit_spread and term_spread reflect the market expectations are also confirmed significantly affect the financial stability. From the perspective of the impact of market expectations on financial stability, on the one hand under the guidance of pessimistic expectations, total demand and total supply reduction, prices and asset prices falling, which leads to rapid macroeconomic contraction; on the other hand, confident optimistic expectations and panic pessimistic expectations will quickly lead to the formation and bursting of asset price bubbles, leading to the financial market turmoil. Therefore, the government and financial supervision departments should increase monetary policy support for the real economy, reduce taxes and fees, ease market access, create a stable, transparent and predictable market environment, maintain a basically stable exchange rate. Through these measures, market expectations will be stabilized and market vitality will be stimulated.

Finally, indicators that reflect the health of the real estate market such as REI_CMAX and REI illiquidity will affect the financial stability. From the perspective of the impact of the real estate market on financial stability, housing price decline rapidly will lead to a series phenomenon such as developers' difficulty survival, banks face supply cuts; housing price rising continuously will lead to the real estate market bubble, which may cause systemic financial risk. Therefore, on the one hand, the government and financial supervision departments must stabilize housing prices, and support residents' reasonable housing needs; on the other hand, it is necessary to resolutely implement the principle of "housing to live without speculation", continue to curb the financialization of real estate market, and maintain the stable and healthy development of the real estate market.

CRedit authorship contribution statement

Benyan Tan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – review & editing, Visualization. **Ziqi Gan:** Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing –

original draft, Writing – review & editing, Visualization. **Yan Wu:** Software, Validation, Investigation, Resources, Data curation, Writing – review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- Acharya, V., Engle, R., & Richardson, M. (2012). Capital shortfall: A new approach to ranking and regulating systemic risks. *American Economic Review*, 102(3), 59–64. <https://doi.org/10.1257/aer.102.3.59>
- Ahmed, N. K., Atiya, A. F., Gayar, N. E., & El-Shishiny, H. (2010). An empirical comparison of machine learning models for time series forecasting. *Econometric Reviews*, 29(5–6), 594–621. <https://doi.org/10.1080/07474938.2010.481556>
- Albulescu, C. T. (2010). Forecasting the Romanian financial system stability using a stochastic simulation model. *Romanian Journal of Economic Forecasting*, 13(1), 81–98. https://ipe.ro/rjef/rjef1_10/rjef1_10_6.pdf
- Alessi, L., & Detken, C. (2018). Identifying excessive credit growth and leverage. *Journal of Financial Stability*, 35, 215–225. <https://doi.org/10.1016/j.jfs.2017.06.005>
- Allen, W. A., & Wood, G. (2006). Defining and achieving financial stability. *Journal of Financial Stability*, 2, 152–172. <https://doi.org/10.1016/j.jfs.2005.10.001>
- Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *Journal of Finance*, 23(4), 589–609.
- Arzamasov, V., & Penikas, H. (2014). A financial stability index for Israel. *Procedia Computer Science*, 31, 985–994. <https://doi.org/10.1016/j.procs.2014.05.351>
- Balakrishnan, R., Danninger, S., Elekdag, S., & Tytell, I. (2011). The transmission of financial stress from advanced to emerging economies. *Emerging Markets Finance and Trade*, 47(sup2), 40–68. <https://doi.org/10.2753/REE1540-496X4703S203>
- Belloni, A., Chernozhukov, V., Fernández-Val, I., & Hansen, C. (2017). Program evaluation and causal inference with high-dimensional data. *Econometrica*, 85(1), 233–298. <https://doi.org/10.3982/ECTA12723>
- Beutel, J., List, S., & von Schweinitz, G. (2019). Does machine learning help us predict banking crises? *Journal of Financial Stability*, 45, Article 100693. <https://doi.org/10.1016/j.jfs.2019.100693>
- Bisias, D., Flood, M., Lo, A. W., & Valavanis, S. (2012). A survey of systemic risk analytics. *Annual Review of Financial Economics*, 4(1), 255–296. <https://doi.org/10.1146/annurev-financial-110311-101754>
- Borio, C. (2003). Towards a macroprudential framework for financial supervision and regulation?. Retrieved from <https://www.bis.org/publ/work128.pdf>
- Brownlees, C., & Engle, R. F. (2017). SRISK: A conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1), 48–79. <https://doi.org/10.1093/rfs/hhw060>
- Bussiere, M., & Fratzscher, M. (2006). Towards a new early warning system of financial crises. *Journal of International Money and Finance*, 25(6), 953–973. <https://doi.org/10.1016/j.jimonfin.2006.07.007>
- Cardarelli, R., Elekdag, S., & Lall, S. (2011). Financial stress and economic contractions. *Journal of Financial Stability*, 7(2), 78–97. <https://doi.org/10.1016/j.jfs.2010.01.005>
- Chen, T., & Guestrin, C. (2016, August). XGBoost: a scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.
- Cipollini, A., & Kapetanios, G. (2009). Forecasting financial crises and contagion in Asia using dynamic factor analysis. *Journal of Empirical Finance*, 16(2), 188–200. <https://doi.org/10.1016/j.jempfin.2008.10.004>
- Davis, E. P., & Karim, D. (2008). Could early warning systems have helped to predict the sub-prime crisis? *National Institute Economic Review*, 206, 35–47. <https://doi.org/10.1177/0027950108099841>
- Deng, C., & Xie, J. (2021). The financial stability and its correlation with business cycle and financial cycle in China (translated from mandarin). *Studies of International Finance*(07), 13–23. <https://doi.org/10.16475/j.cnki.1006-1029.2021.07.002>
- Duprey, T., Klaus, B., & Peltonen, T. (2017). Dating systemic financial stress episodes in the EU countries. *Journal of Financial Stability*, 32, 30–56. <https://doi.org/10.1016/j.jfs.2017.07.004>
- Erdogan, O., Bennett, P. B., & Ozyildirim, C. (2012). An early warning signal of financial crisis by using the deepness and liquidity in stock markets. *International Review of Applied Financial Issues and Economics*, 4(1). <https://www.researchgate.net/publication/255967569>
- Fornaro, P., & Luomaranta, H. (2020). Nowcasting Finnish real economic activity: A machine learning approach. *Empirical Economics*, 58(1), 55–71. <https://doi.org/10.1007/s00181-019-01809-y>
- Frankel, J. A., & Rose, A. K. (1996). Currency crashes in emerging markets: An empirical treatment. *Journal of International Economics*, 41(3–4), 351–366. [https://doi.org/10.1016/S0022-1996\(96\)01441-9](https://doi.org/10.1016/S0022-1996(96)01441-9)
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of statistics*, 1189–1232. <https://www.jstor.org/stable/2699986>
- Gersl, A., & Hermanek, J. (2007). Financial stability indicators: Advantages and disadvantages of their use in the assessment of financial system stability. Retrieved from. <https://ideas.repec.org/h/cnb/ocpubc/fsr06-2.html>
- Gogas, P., Papadimitriou, T., & Agravetidou, A. (2018). Forecasting bank failures and stress testing: A machine learning approach. *International Journal of Forecasting*, 34(3), 440–455. <https://doi.org/10.1016/j.ijforecast.2018.01.009>
- Goldstein, A., Kapelner, A., Bleich, J., & Pitkin, E. (2015). Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation. *Journal of Computational and Graphical Statistics*, 24(1), 44–65. <https://doi.org/10.1080/10618600.2014.907095>
- Goodhart, C., & Hofmann, B. (2001, March). *Asset prices, financial conditions, and the transmission of monetary policy*. In Proceedings of the Conference on Asset Prices, Exchange Rates, and Monetary Policy, Stanford University.
- Goulet Coulombe, P., Leroux, M., Stevanovic, D., & Surprenant, S. (2022). How is machine learning useful for macroeconomic forecasting? *Journal of Applied Econometrics*, 37(5), 920–964. <https://doi.org/10.1002/jae.2910>
- Grabel, I. (2000). Identifying risks, preventing crisis: Lessons from the Asian crisis. *Journal of Economic Issues*, 34(2), 377–383. <https://www.jstor.org/stable/4227566>
- Gramegna, A., & Giudici, P. (2021). SHAP and LIME: An evaluation of discriminative power in credit risk. *Frontiers in Artificial Intelligence*, 4, Article 752558. <https://doi.org/10.3389/frai.2021.752558>
- Hakkio, C. S., & Keeton, W. R. (2009). *Financial stress: What is it, how can it be measured, and why does it matter*. Retrieved from <https://www.kansascityfed.org>
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 357–384. <https://doi.org/10.2307/1912559>
- Hodrick, R. J., & Prescott, E. C. (1997). Postwar US business cycles: An empirical investigation. *Journal of Money, Credit, and Banking*, 29(1), 1–16. <https://doi.org/10.2307/2953682>
- Hollo, D., Kremer, M., & Lo Duca, M. (2012). *CISS-a composite indicator of systemic stress in the financial system*. ECB Working Paper, No.1426. <https://www.econstor.eu/handle/10419/153859>
- Holopainen, M., & Sarlin, P. (2017). Toward robust early-warning models: A horse race, ensembles and model uncertainty. *Quantitative Finance*, 17(12), 1933–1963. <https://doi.org/10.1080/14697688.2017.1357972>
- Houben, A. G., Kakes, J., & Schinas, G. J. (2004). Toward a framework for safeguarding financial stability. *IMF Working Papers*, 2004(101). <https://doi.org/10.5089/9781451852547.001>
- Hu, H., Tang, L., Zhang, S., & Wang, H. (2018). Predicting the direction of stock markets using optimized neural networks with Google Trends. *Neurocomputing*, 285, 188–195. <https://doi.org/10.1016/j.neucom.2018.01.038>
- Illing, M., & Liu, Y. (2006). Measuring financial stress in a developed country: An application to Canada. *Journal of Financial Stability*, 2(3), 243–265. <https://doi.org/10.1016/j.jfs.2006.06.002>
- Ivanková, K. (2012). *Financial stability indicator predictability by support vector machines*. In Proceedings of 30th International Conference Mathematical Methods in Economics. <http://mme2012.opf.slu.cz/proceedings/pdf/062.Ivankova.pdf>
- Joseph, A. (2019a). *A framework for statistical inference on machine learning models*. Retrieved from <https://www.bankofengland.co.uk/working-paper/staff-working-papers>
- Joseph, A. (2019b). *Parametric inference with universal function approximators*. Retrieved from <https://doi.org/10.48550/arXiv.1903.04209>
- Kaminsky, G., Lizondo, S., & Reinhart, C. M. (1998). Leading indicators of currency crises. *Staff Papers*, 45(1), 1–48. <https://link.springer.53yu.com/article/10.2307/3867328>
- Kaufman, G. G., & Scott, K. E. (2003). What is systemic risk, and do bank regulators retard or contribute to it? *The Independent Review*, 7(3), 371–391. <https://www.jstor.org/stable/24562449>
- Kliesen, K. L., & Smith, D. C. (2010). Measuring financial market stress. *economic synopses*.
- Koong, S. S., Law, S. H., & Ibrahim, M. H. (2017). Credit expansion and financial stability in Malaysia. *Economic Modelling*, 61, 339–350. <https://doi.org/10.1016/j.econmod.2016.10.013>
- Li, H., Cao, Y., Li, S., Zhao, J., & Sun, Y. (2020). XGBoost model and its application to personal credit evaluation. *IEEE Intelligent Systems*, 35(3), 52–61. <https://doi.org/10.1109/MIS.2020.2972533>
- Li, M., & Liang, S. (2021). Monitoring systemic financial risks: Construction and state identification of China's financial market stress index (translated from mandarin). *Journal of Financial Research*, 49(2), 21–38.
- Lundberg, S. M., & Lee, S. I. (2017). *A unified approach to interpreting model predictions*. In proceedings of 31st Conference on Neural Information Processing Systems (NIPS 2017).
- Ma, X., Sha, J., Wang, D., Yu, Y., Yang, Q., & Niu, X. (2018). Study on a prediction of P2P network loan default based on the machine learning LightGBM and XGboost algorithms according to different high dimensional data cleaning. *Electronic Commerce Research and Applications*, 31, 24–39. <https://doi.org/10.1016/j.elerap.2018.08.002>
- Matheson, T. D. (2012). Financial conditions indexes for the United States and euro area. *Economics Letters*, 115(3), 441–446. <https://doi.org/10.1016/j.econlet.2011.12.119>
- Mishkin, F. S. (2011). Over the cliff: From the subprime to the global financial crisis. *Journal of Economic Perspectives*, 25(1), 49–70. <https://doi.org/10.1257/jep.25.1.49>
- Morales, M. A., & Estrada, D. (2010). A financial stability index for Colombia. *Annals of Finance*, 6(4), 555–581. <https://doi.org/10.1007/s10436-010-0161-7>

- Nag, A., & Mitra, A. (1999). Neural networks and early warning indicators of currency crisis. *Reserve Bank of India Occasional Papers*, 20(2), 183–222. <https://www.researchgate.net/publication/284480973>.
- Petrovska, M., & Mihajlovska, E. M. (2013). Measures of financial stability in Macedonia. *Journal of central banking Theory and practice*, 2(3), 85–110. <https://EconPapers.repec.org/RePEc:cbk:journl:v:2:y:2013:i:3:p:85-110>.
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why should i trust you?” Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining.
- Sachs, J. D., Tornell, A., & Velasco, A. (1996). Financial crises in emerging markets: The lessons from 1995. *NBER Working Paper Series*, No.5576. <https://www.nber.org/papers/w5576>.
- Saisana, M., & Tarantola, S. (2002). *State-of-the-art report on current methodologies and practices for composite indicator development*. <https://www.researchgate.net/publication/305392511>.
- Samitas, A., Kampouris, E., & Kenourgios, D. (2020). Machine learning as an early warning system to predict financial crisis. *International Review of Financial Analysis*, 71. <https://doi.org/10.1016/j.irfa.2020.101507>
- Schinasi, G. J. (2004). *Defining financial stability*. <https://doi.org/10.5089/9781451859546.001>
- Shapley, L. S. (1953). A Value for N-Person Games. *Contributions to the Theory of Games*, 307–317.
- Tanaka, K., Kinkyo, T., & Hamori, S. (2016). Random forests-based early warning system for bank failures. *Economics Letters*, 148, 118–121. <https://doi.org/10.1016/j.econlet.2016.09.024>
- Varian, H. R. (2014). Big data: New tricks for econometrics. *Journal of Economic Perspectives*, 28(2), 3–28. <https://doi.org/10.1257/jep.28.2.3>
- Wheelock, D. C., & Wohar, M. E. (2009). Can the term spread predict output growth and recessions? A survey of the literature. *Federal Reserve Bank of St. Louis Review*, 91(5 Part 1), 419–440.
- Xu, G., & Li, B. (2017). Study on the construction of China’s financial stress index and its dynamic transmission effect (translated from mandarin). *Statistical Research*, 34(4), 59–71. <https://doi.org/10.19343/j.cnki.11-1302/c.2017.04.006>
- Yiu, M. S., Ho, W. Y. A., & Jin, L. (2010). A measure of financial stress in Hong Kong financial market—the financial stress index. *Hong Kong Monetary Authority Research Note*, 2, 2010.
- Zhang, J., & Gao, Q. (2015). A study on the design and applications of China’s financial systemic stress index (translated from mandarin). *Journal of Quantitative & Technological Economics*, 32(11), 41–57. <https://doi.org/10.13653/j.cnki.jqte.2015.11.003>
- Zhao, Q., & Hastie, T. (2021). Causal interpretations of black-box models. *Journal of Business & Economic Statistics*, 39(1), 272–281. <https://doi.org/10.1080/07350015.2019.1624293>
- Zhou, J., Li, W., Wang, J., Ding, S., & Xia, C. (2019). Default prediction in P2P lending from high-dimensional data based on machine learning. *Physica A: Statistical Mechanics and its Applications*, 534, Article 122370. <https://doi.org/10.13653/j.cnki.jqte.2015.11.003>